

Linear Optimizat'



Date 30 / 7 / 2025

L-02

$$A = \begin{matrix} & \begin{matrix} x_1 & x_2 & x_3 & \dots & x_n \end{matrix} \\ \begin{matrix} R_1 \\ R_2 \\ \vdots \\ R_m \end{matrix} & \begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \end{matrix} \quad B = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix}$$

- (1) Row exchange $R_i \leftrightarrow R_j$ } solⁿ remains same
 (2) $R_i \leftarrow R_i + \alpha R_j$ } space $\{x_1, x_2, \dots, x_n\}$

Rotating line i with line j without changing Intersecⁿ point

$$A \rightarrow A' ; B \rightarrow B'$$

$$Ax = B, \quad A'x = B'$$

Not necessary (with variables)

- (3) $C_i \leftrightarrow C_j$ Column Exchange

Don't care about it

$$\begin{array}{c} \text{Identity} \\ \left\{ \begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & a_{1s} & \dots & a_{1n} \\ 0 & 1 & 0 & 0 & a_{2s} & \dots & a_{2n} \\ 0 & 0 & 1 & 0 & a_{3s} & \dots & a_{3n} \\ 0 & 0 & 0 & 1 & a_{4s} & \dots & a_{4n} \\ \hline 0 & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & 0 & \dots & 0 \end{array} \right\} \begin{bmatrix} b_1' \\ b_2' \\ b_3' \\ b_4' \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \end{array}$$

zero

zero

If any non zero we don't have solⁿ

$$x_1 = b_1', \quad x_2 = b_2', \quad \dots, \quad x_n = b_n'$$

$$\text{Solⁿ set } S = \{x : Ax = 0\}$$

Abelian group : Associativity Inverse
 Identity Commutativity



L-3

Vector space

→ all properties (Requirements)

$$v_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \quad v_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad v_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \quad v_4 = \begin{bmatrix} 3 \\ 2 \\ 5 \end{bmatrix}$$

$$v_4 = 3v_1 + 2v_2 + 5v_3$$

$$v_1 \quad v_2 \quad v_3 \quad \dots \quad v_n \quad w \quad \text{in } V$$

$$w = \alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 + \dots + \alpha_n v_n$$

$$-\infty < \alpha_i < \infty$$

$$S = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\} \quad \text{Basis of } V = \{(x, y, z)\}$$

↳ Multiple

↳ Linear Independent

Imp → Any 2 Basis for a given vector space, cardinality of any 2 basis remains same (Proof by contradiction)

Given a V & Basis B :

$$S = \{v_1, v_2, v_3, \dots, v_n\} \quad w \in V$$

$$w = \alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_n v_n$$

$$w = \beta_1 v_1 + \beta_2 v_2 + \dots + \beta_n v_n$$

Any vector in V can only expressed in a single way.

$$0 = v_1(\alpha_1 - \beta_1) + v_2(\alpha_2 - \beta_2) + \dots + v_n(\alpha_n - \beta_n)$$

⇒ Proof By Contradict"



Date ____ / ____ / ____

Basis

due to
Proof

Given vector space V , $\|A\| = m$ & $\|B\| = n$
 $m < n$

$A = \{w_1, w_2, \dots, w_k, a_{k+1}, a_{k+2}, \dots, a_m\}$ k common elements
 $B = \{w_1, w_2, \dots, w_k, b_{k+1}, \dots, b_n\}$

$k = n \Rightarrow$ Not possible as then B is proper subset of A .

Then A is not basis as it won't be linearly independent

thm
Lec 4

$A = w_1, w_2, w_3, \dots, w_k, a_{k+1}, a_{k+2}, \dots, a_m$

$B = w_1, w_2, w_3, \dots, w_k, b_{k+1}, b_{k+2}, \dots, b_n$

$$b_{k+1} = \alpha_1 w_1 + \alpha_2 w_2 + \dots + \alpha_k w_k + \alpha_{k+1} a_{k+1} + \dots + \alpha_m a_m$$

$$\alpha_{k+1} = 0, \alpha_{k+2} = 0, \dots, \alpha_m = 0 \quad \text{Not possible}$$

otherwise B is not linearly independent

$\alpha_{k+1} \neq 0$
without loss of generality

Now $A' = A - a_{k+1} + b_{k+1}$

$$b_{k+1} = \beta_1 w_1 + \beta_2 w_2 + \dots + \beta_k w_k + \beta_{k+2} a_{k+2} + \dots + \beta_n a_n$$

$$\alpha_{k+1} = \gamma_1 w_1 + \gamma_2 w_2 + \dots + \gamma_k w_k + \gamma_{k+1} a_{k+1} + \dots + \gamma_n a_n$$

$$A \quad S = \{x : Ax = 0\}$$

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ a_{31} & a_{32} & \dots & a_{3n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix} \quad m \times n$$

Date / /

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

No. of linearly independent
col = No. of linearly independent
row = rank of matrix
By Gauss elimination

6th Aug, 2022

Lec-5

No. of LD Rows = No. of LD Columns

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

$R_i \leftrightarrow R_j$ } Doesn't change Rank
 $C_i \leftrightarrow C_j$ } Row space doesn't change

$$R_i \leftarrow R_i + \alpha R_j$$

$$C_i \leftarrow C_i + \alpha C_j$$

$A \rightarrow A'$ Independent Rows (A') = IR (A')
Row operatⁿ $\nearrow$
 A' (Row space)

- * Row operⁿ doesn't change linearly independent Rows
- * Column space might change, max no. of independent Col remains same.

$$S = \{x : Ax = 0\} \quad \text{columns} \quad m \times n \quad \text{Rank}(A) = k$$

Rank Nullity Theorem

$$\boxed{n - k} \quad A_{m \times n} \quad k \leq \min(m, n)$$

7th Aug

$$S = \{x : Ax = 0\} \quad \text{can be derived?}$$

$$S' = \{x : Ax = b\} \quad \text{shifting by some constant } b$$

$$x + y \leq 2$$



Date ____ / ____ / ____

11th Aug, 2023

L-7

$A_{m \times n}$

B_m

Convex Set

$$S = \{x : Ax \leq B\}$$

$$u, v \in S \quad 0 \leq \alpha \leq 1 \Rightarrow \alpha u + (1-\alpha)v \in S$$

Convex Combinatⁿ

Proof

$$AU \leq B$$

$$AV \leq B$$

$$A(\alpha u) \leq \alpha B$$

$$A(1-\alpha)v \leq (1-\alpha)B$$

Vertex of Set



$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

$$S = \{Ax \leq B\}$$

3×2

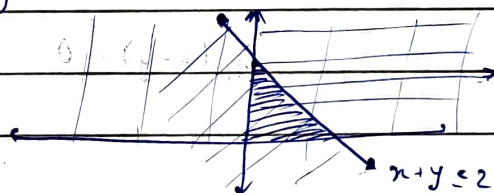
Vertex $\rightarrow$ In n dim.

n planes will intersect

$\Rightarrow$ Degenerate System

$\rightarrow$ In n dim, $n+1$ intersection (Vertex)

Tight / Untight



13th Aug, 2023

(1-8) No. of linearly independent row = Rank of matrix

Let x be a vertex

Let A_i is tight; $A_i \rightarrow$ untight; Rank $(A_i) = n$

If x is vertex; Rank $(A_i) = r$; if Rank (A_i) is n then x is a vertex

$\Rightarrow$ If Rank $(A_i) < n$; then x lies on line (prove by contradiction)

$$S = \{x : Ax = B\}$$

$$x \quad x \in S$$

$$x = z + \alpha u \quad \alpha \geq 0$$

$$x = \frac{x+y}{2} \text{ (not vertex)}$$

$$y = z - \alpha u$$

Date ___/___/___



14th Aug, 2025

$$\text{Let } S = \{x : Ax \leq B\}$$

$x \in S$ If $\text{Rank}(A) = n$ then x is a vertex.

⇒ Proof by contradiction

$$x, y \quad z = \alpha x + (1-\alpha)y \quad 0 < \alpha < 1$$

$$A_1 x = b_1 \dots b_k$$

$$A_1 y = b_1 \dots b_k$$

$$A_1 z = b_1 \dots b_k$$

$$A_1 (x-y) = 0 \quad \text{Not linearly independent}$$

$$\text{Rank} < n$$

Local max = Global max Convex f^*

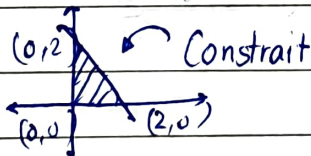
18th Aug, 2025

$$C = 2x + 3y$$

We have to check

on vertex point.

(That'll be optimal pt.)



$$S = \{x : Ax \leq B\}$$

$z \in S$ is an optimum point.

$z \rightarrow z' \rightarrow z'' \rightarrow \dots$ (vertex) while loop

Increase the tight row ↗

Assume $A_{m \times n}$ $\text{Rank}(A) = n$

At z
 ↗ tight rows A_1
 ↘ Untight rows A_2

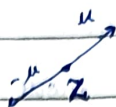
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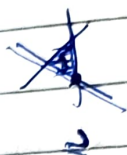
$$\exists U \in \mathbb{R}^n$$

$$\therefore A, U = 0 \quad \text{Rank}(A) < n$$

Reach towards a vertex:



$CU \neq 0$? No, it has to be zero.



20th Aug

Optimum is at a vertex.



Let A_1 be tight rows; A_2 be unlight rows

while $(\text{Rank}(A_1) < n)$

$$Z = z + \alpha U$$

$$\therefore \text{Let } U \in \mathbb{R}^n \quad A, U = 0$$

$C, U = 0$ o/w Z is not a optimum.

$A_2, U \neq 0$ o/w $\text{Rank}(A) < n$.

$$\therefore Z' = z + \alpha U \quad \{ A_1 \text{ remains tight}$$

at least one row of A_1 is tight

& objective remains same. }

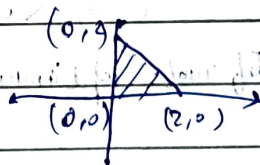
$A_{m \times n}$ at $\text{Rank}(n)$, B_m , C_n

(*)

$$Ax \leq B \quad ; \quad \max C^T x$$

Simplex algorithm :- Jump from one vertex to other.
where objective might increase/decrease

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$



Unbounded v/s Bounded

$A_{m \times n}$, B_m , C_n

Find $x \in \mathbb{R}^n$; $Ax \leq B$; $\max C^T x$

①

$\text{Rank}(A) = n$ ④ some one given vertex z .

②

$S = \{x; Ax \leq B\}$ is bounded.

③ S : S is non-degenerate.

Date ___/___/___ now?



while (z is not a optimum vertex)

{ move to a neighbour vertex with better objective f^* }

25th Aug

If any matrix having n ($n \times n$), there exists a matrix for the inverse.

Simpler

Let y be a optimum vertex

Algorithm

$$U = Y - Z \quad (A_1 Y - A_1 Z \leq 0)$$

$$A_1 U \leq 0$$

$$C^T U > 0$$

$$C^T Y > C^T Z$$

$$A = \begin{bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{bmatrix}$$

$$U = \alpha_1 b_1 + \alpha_2 b_2 + \dots + \alpha_n b_n$$

$$\begin{bmatrix} | & | & | & | & | \\ b_1 & b_2 & & & b_n \end{bmatrix}$$

$$A_1 U = A_1 (\alpha_1 b_1 + \alpha_2 b_2 + \dots + \alpha_n b_n)$$

All the α_i are at most zeroes.

Inverse of A

$$C^T U = C^T (\alpha_1 b_1 + \alpha_2 b_2 + \dots + \alpha_n b_n)$$

$$C^T (\alpha, b_i) > 0$$

z is not optimum

$$C^T b_i < 0$$

If not optimum

28th Aug

$A_{m \times n}$, B_m , C_n

$$\text{Rank}(A) = n$$

$S \{x : Ax \leq B\}$ Bounded non degenerate

someone given initial vertex z

$$\begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix}$$

$$\begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix}$$

$$b_1, b_2, b_3 \geq 0$$

Input $\rightarrow A, B, C$
4 assumpt.

$\rightarrow$ Numpy

$\rightarrow$ sequence of jumps that's making objective $\uparrow$

$\rightarrow$ Compute Truss

Date $___ / ___ / ___$

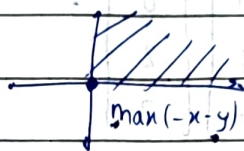


$$\begin{bmatrix} x & y \\ -1 & 0 \\ 0 & -1 \\ 1 & 1 \end{bmatrix}$$

$x_1 + 11x_2 + 0y_1 - 0y_2 \rightarrow$ Replace a variable with 2 variable (non-neg.)

$$(x_1 - x_2) - (y_1 - y_2)$$

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$



$\max(-x-y)$ Point (Bounded)

$\max(x+y)$ Unbounded

1st step

$A_{m \times n} \quad B_m \quad C_n \quad \{ \text{system is not bounded} \}$

$\circ$ A is non degenerate

$\circ$ someone given feasible pt. x

$\Rightarrow$ Divide tight & untight. Increase the tight rows

A_1, A_2

$U \in \mathbb{R}^n \quad A_1 U = 0 \quad \{ \text{Dirac}^n \text{ of } U \mid \text{opt of } U \}$

Attack on them will increase tight rows

$A_2 U \leq 0 \quad A_1$ Tight But non of A_2 is becoming tight, so it'll

$A_2(-U) \geq 0$ at max new

All entries have to be new (if both to grow) $A_2 \neq 0$

$AU \leq 0$ Rank is n come to vertex

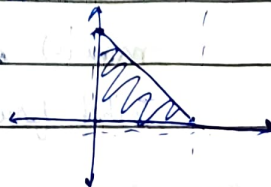
Diracⁿ vector we calculate by inverse of matrix

$A_{m \times n} \quad B_m \quad C_n$

$\circ$ same one given feasible pt. x

$$A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 2 \\ 2 \end{bmatrix} + \epsilon$$



lines are accepted slightly

$\epsilon \approx 0$

$\exists \epsilon$ system becomes non-degenerate

Date ____ / ____ / ____

8th Sept

$A_{m \times n}$ B_m C_n

○ Same given initial feasible pt. x .

$$\lambda t \approx 0$$

$$\begin{bmatrix} -1 & 0 \\ 0 & -1 \\ 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 0+t \\ 0+t^2 \\ 2+t^3 \\ 2+t^4 \end{bmatrix}$$

Vertex $\rightarrow$ May split into many vertex
Degeneracy will go.

○ If after changing unbounded; original should also be unbounded.

$\rightarrow$ Take ϵ ; if still degenerate $\epsilon/2$; $\epsilon/2^2$... until system becomes degenerate.

★ $A_{m \times n}$ B_m C_n

○ Increase dimen^s, to get system easy to solve

$$\begin{matrix} x & y & z \end{matrix}$$

$$\begin{bmatrix} -1 & 0 & 1 \\ 0 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 2 \\ 2 \\ -\min(z) \end{bmatrix} \quad \text{maximizing } (z)$$

$\max(z)$ should be non-negative; for the feasibility.
Initial feasible pt $\rightarrow (x, y, z)$



$$x^2 = 1$$

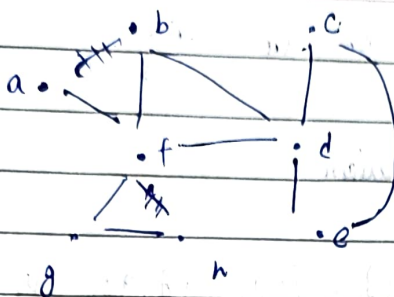
$$O((11V+11V) \log 11V) \quad 8^{+8}$$



Date ____ / ____ / ____

Vertex Cover Problem

10th Sept 2025



Graph: Input

Output: Where does each point; Minimise it so all the edges are covered.

GRAPH Isomorphism

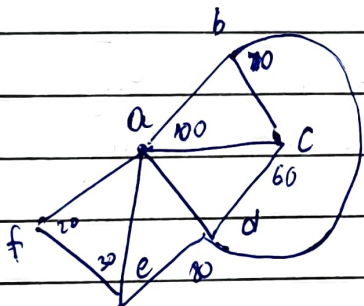
2-approximative algorithm

⇒ "approximative" ⇒ 2^{approx}

⇒ Think for a 2-approximative algorithm

(a,b) (f,b) (d,e)

2-approximative algorithm



Integer Linear Programming

Longitude & Latitude

$$x_a, x_b, x_c, x_d, x_e, x_f \in \{0, 1\}$$

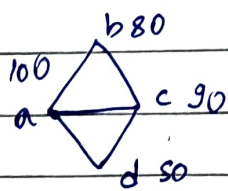
$$100x_a + 80x_b + 60x_c + 80x_d + 30x_e + 20x_f$$

Objective F"

$$\left. \begin{aligned} x_a + x_b &\geq 1 \\ x_b + x_c &\geq 1 \\ x_a + x_c &\geq 1 \end{aligned} \right\} \text{Constraints}$$

Relax This

11th Sept



$$\min \{ x_a, x_b, x_c, x_d \in \{0, 1\} \}$$

$$(100x_a + 80x_b + 90x_c + 50x_d)$$

$$x_a + x_b \geq 1$$

$$x_b + x_c \geq 1$$

$$x_a, x_b, x_c, x_d \geq 0$$

$$x_c + x_d \geq 1$$

$$x_a + x_d \geq 1, x_a + x_c \geq 1$$

opt* Sopt* Tampering to go to opt*, shouldn't change more than twice (significant)

Sopt* ≤ opt*



Date Randomize

$$x_a', x_b', x_c', x_d' < 0, s = 0$$

$$> 0, s = 1$$

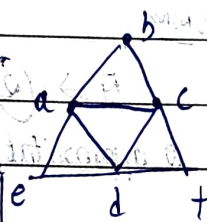
Randomized
algorithm
Using
Randomizing

det - Linear problem

$$\{ \{a, b\}, \{b, c\}, \{c, d\}, \{d, a\}, \{a, c\} \}$$

$$\{a, b, c, d\}$$

$$100 \quad 60 \quad 90 \quad 60$$



$((a, b), (d, e))$

Primal space & Dual Problem

15th Sept, 2023

$$\begin{bmatrix} s_1 & s_2 & s_3 & s_n \end{bmatrix}$$

$$\begin{bmatrix} c_1 & c_2 & c_3 & c_n \end{bmatrix}$$

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ \vdots & \vdots \\ a_{m1} & a_{m2} \end{bmatrix}$$

what if we ~~do~~ sell directly

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_m \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix}$$

$$(x_1, x_2, x_3, \dots, x_n) \dots x_1, \dots, x_n \geq 0$$

$$\max (c_1 x_1 + c_2 x_2 + \dots + c_n x_n)$$

$$\min (b_1 y_1 + b_2 y_2 + \dots + b_m y_m) \quad y_1 \geq 0, y_2 \geq 0, \dots, y_m \geq 0$$

(Purchase)

$$a_{11} y_1 + a_{21} y_2 + a_{31} y_3 + \dots + a_{m1} y_m \geq c_1 \quad (\text{Directly sell})$$

$$a_{1n} y_1 + a_{2n} y_2 + a_{3n} y_3 + \dots + a_{mn} y_m \geq c_n$$

Primal $\rightarrow$ Dual
Variable $\rightarrow$ Constant
Constant $\rightarrow$ Variable

Date ____ / ____ / ____



max

(2+)

(y+)

Row

Optimal pts

Multiply y_1 (after x_1); 'Eq' will come automatically.

Primal feasible $\Rightarrow$ Dual Feasible
is at most of

22nd Sept, 2025

Primal $\rightarrow$ Dual

$$\begin{array}{c}
 C_1 x_1 \quad C_2 x_2 \quad C_3 x_3 \quad \dots \quad C_n x_n \\
 y_1 \left[\begin{array}{cccc} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \end{array} \right] \leq \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_m \end{array} \right] \\
 y_2 \left[\begin{array}{cccc} a_{21} & a_{22} & a_{23} & \dots & a_{2n} \end{array} \right] \leq \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_m \end{array} \right] \\
 \vdots \\
 y_m \left[\begin{array}{cccc} a_{m1} & a_{m2} & a_{m3} & \dots & a_{mn} \end{array} \right] \leq \left[\begin{array}{c} b_1 \\ b_2 \\ \vdots \\ b_m \end{array} \right]
 \end{array}$$

First multiply by $x_1, x_2, \dots$
Then by $y_1, y_2, \dots$

$$\left[\begin{array}{cccc} -1 & 0 & 0 & \dots & 0 \\ 0 & -1 & 0 & \dots & 0 \\ 0 & 0 & 0 & \dots & -1 \end{array} \right] \left[\begin{array}{c} 0 \\ 0 \\ 0 \end{array} \right]$$

$b_1 \quad b_2 \quad b_m$

Dual Constraint

$$\left[\begin{array}{l} a_{11} y_1 + a_{21} y_2 + \dots + a_{m1} y_m \geq C_1 \\ a_{12} y_1 + a_{22} y_2 + \dots + a_{m2} y_m \geq C_2 \\ \vdots \\ a_{1n} y_1 + a_{2n} y_2 + \dots + a_{mn} y_m \geq C_n \end{array} \right]$$

$$\left[\begin{array}{cccc} 1 & 0 & & 0 \\ 0 & 1 & & 0 \\ & & \ddots & \\ 0 & 0 & & 1 \end{array} \right] \left[\begin{array}{c} 0 \\ 0 \\ 0 \\ 0 \end{array} \right]$$

Primal	Dual
$x_1, x_2, \dots, x_n$	$y_1, y_2, \dots, y_m$
$C_1 x_1 + C_2 x_2 + \dots + C_n x_n \leq$	$b_1 y_1 + b_2 y_2 + \dots + b_m y_m$

Everything is non-negative, otherwise proof will not hold.

Date ____ / ____ / ____



Primal Opt = Dual Opt $\Rightarrow$ Non Degenerate

$\nearrow$ cost

A, c, b
 $m \times n \quad n \quad m$

Tight

A_1, A_2

$\rightarrow$ Divide tight & untight rows $(n \times n) \quad (m-n, n)$

Rows of $A_1 \rightarrow R_1, R_2, \dots, R_n$

K.K.T Cond'

$$\alpha_1 R_1 + \alpha_2 R_2 + \dots + \alpha_n R_n = c, \quad \alpha_i \geq 0$$

Linear combination tight Rows

at optimum pt.
 the cost vector c is

Cost
Proof

If α is negative $x_1, x_2, \dots, x_n$ is not optimal

non linear
 comb of
 Tight Rows
 (not)

Let $D = -\text{inv}(A_1)$ $\begin{bmatrix} 1 & | & | & | \end{bmatrix}$

$\hookrightarrow$ Will increase Objective f^*

Move in Direcⁿ of D